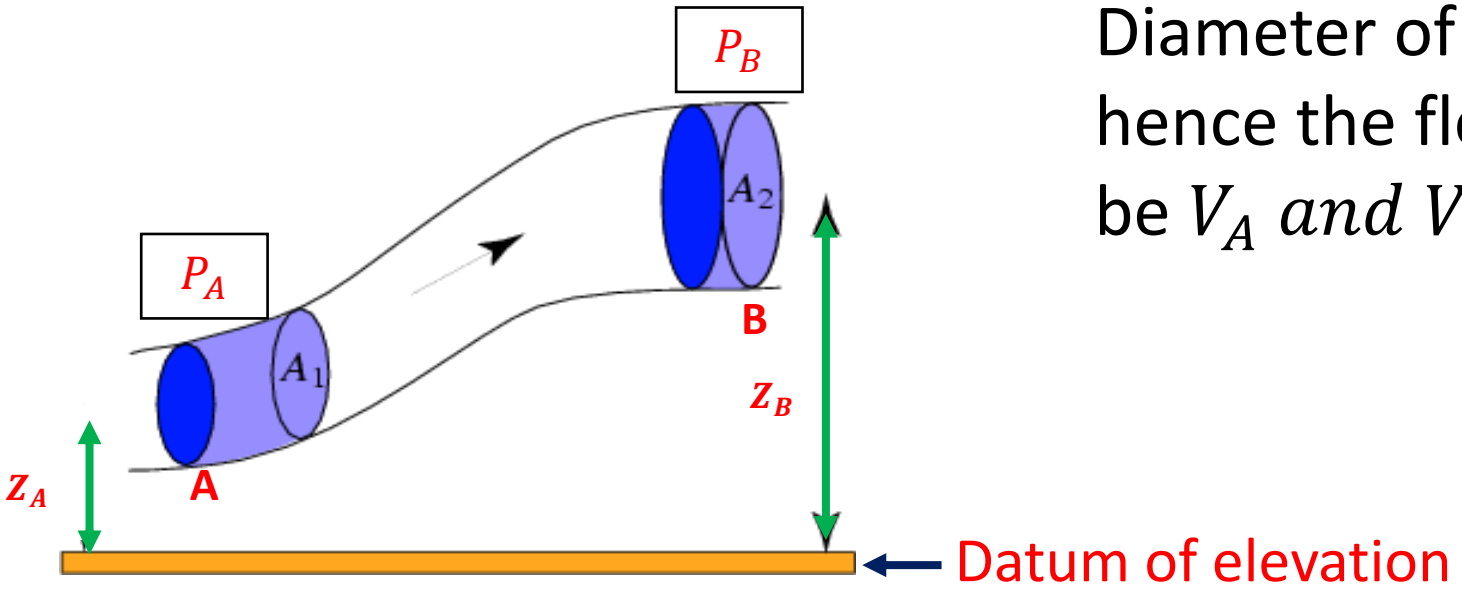
Gas Flow Equation

Instructor: Prof. Vinay Kumar Rajak

Bernoulli's Equation

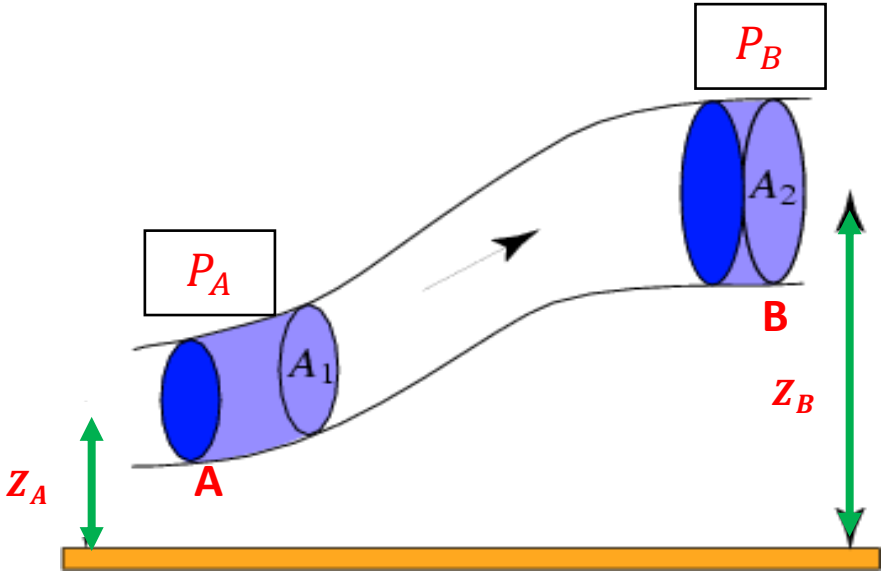


Diameter of A and B are different hence the flow velocity at A & B will be V_A *and* V_B

Principle of conservation of energy is applied to liquid flow through a pipeline.

A particle of the liquid of unit weight (also known as specific weight, i.e. weight per unit volume of a material) at any point 'A' in the pipeline possesses a total energy 'E', which consists of 3 components.

Bernoulli's Equation



Datum of elevation

Potential Energy (per unit weight) = Z_A

Pressure Energy (per unit weight) = $\frac{P_A}{\rho}$

Kinetic Energy (per unit weight) = $\frac{{V_A}^2}{2g}$

Where ρ *is specific wt. of liquid in* $\frac{lbf}{ft^3}$

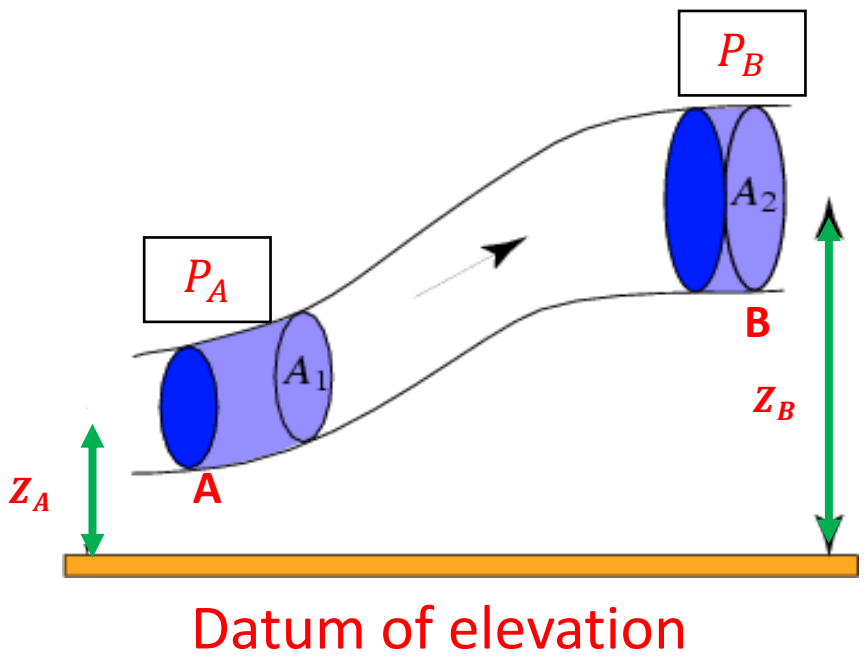
Total Energy E = $Z_A + \frac{P_A}{\rho} + \frac{{V_A}^2}{2g}$

Bernoulli's Equation

Parameters and their Units (FPS)

Parameter	Meaning	Unit (FPS)
Z_A	Elevation (height)	ft
P_A	Pressure	lbf/ft^2
ρ	Mass density	$\frac{lbf}{ft^3}$ or lbm/ft^3
g	Acceleration due to gravity	ft/s^2
V_A	Velocity	ft/s

Bernoulli's Equation



Total Energy at point A is assumed to be H_A in feet of liquid head

$$H_A = Z_A + \frac{P_A}{\rho} + \frac{{V_A}^2}{2g}$$

Similarly at point B

$$H_B = Z_B + \frac{P_B}{\rho} + \frac{{V_B}^2}{2g}$$

By principle of conservation of energy

$$H_A = H_B$$

Bernoulli's Equation

$$Z_A + \frac{P_A}{\rho} + \frac{V_A^2}{2g} = Z_B + \frac{P_B}{\rho} + \frac{V_B^2}{2g}$$

Taking into account the addition of energy (such as from a pump at A) and accounting for frictional head losses h_f , we get the more common form of Bernoulli's equation as follows:

$$Z_A + \frac{P_A}{\rho} + \frac{V_A^2}{2g} + \boldsymbol{H_p} = Z_B + \frac{P_B}{\rho} + \frac{V_B^2}{2g} + \boldsymbol{h_f}$$

$H_p \rightarrow$ Equivalent head added to the liquid by the pump at A and

$h_f \rightarrow$ represents the total frictional head losses between points A and B

$$\frac{P_B}{\rho} - \frac{P_A}{\rho} + h_f = 0$$

$$\frac{dP}{\rho} + \frac{flv^2}{2gd} = 0 \text{ ---------- } (i)$$

Assumptions:

Horizontal, Steady state $\left(\frac{\delta}{\delta t} = 0\right)$, Adiabatic $(Q = 0)$,

Isothermal flow $(T = Contant)$ with negligible kinetic energy change

Real Gas Equation

$PV = ZnRT$ (Where Z= compressibility factor and n= number of lb-moles)

Also,

$$n = \frac{m}{M}, \rho = \frac{m}{V}, \rho = \frac{PM}{ZRT} \rightarrow m = \frac{PVM}{ZRT} \rightarrow \dot{m} = \frac{PQM}{ZRT}$$

Flow Relation

$$\frac{P_{sc}\, Q_{sc}}{T_{sc}\, Z_{sc}} = \frac{P\, Q}{T\, Z}$$

For standard conditions:

$Z_{sc} = 1$

Final Expression for Flow Rate

$$Q = \frac{P_{sc}\, Q_{sc}}{T_{sc}} \times \frac{TZ}{P}$$

$$AV = \frac{P_{sc}\, Q_{sc}}{T_{sc}} \times \frac{TZ}{P}$$

$$\frac{\pi d^2}{4} V = \frac{P_{sc}\, Q_{sc}}{T_{sc}} \times \frac{TZ}{P}$$

$$V = \frac{4P_{sc}}{P}\ \frac{T}{T_{sc}}\ \frac{Q_{sc}Z}{\pi d^2} \quad \text{----------} \ (\boldsymbol{ii})$$

Note: T is constant since isothermal flow is assumed. Otherwise an average temperature T_{av}can be taken.

Either

$$\boldsymbol{T_{av} = \frac{T_1 + T_2}{2}}$$

or

$$\boldsymbol{T_{av} = \frac{T_1 - T_2}{\ln\left(\frac{T_1}{T_2}\right)}}$$

Substituting the value of V in equation (i)

$$-dp = \left(\frac{f}{2gd}\right)\left(\frac{PM}{ZRT}\right)\left(\frac{16Q_{sc}^2 Z^2 T^2 P_{sc}^2}{P^2 T_{sc}^2 \pi^2 d^4}\right) dl$$

$$\Rightarrow -\int \frac{P}{Z}\, dp = \frac{8fMT\, P_{sc}^2 Q_{sc}^2}{R\, T_{sc}^2\, \pi^2 d^5\, g} \int dl \quad ---------- -(iii)$$

Solving the equation with Boundary conditions as

$$L = 0 \;\Rightarrow\; P = P_1$$
$$\mathrm{l} = L \;\Rightarrow\; P = P_2$$

$T = T_f$

where

T_f =flowing temperature (i.e., temperature of fluid flowing in the pipe).

$Z = Z_{av}$

where

Z_{av} =average compressibility factor.

To find the average pressure in the pipeline, using $P_{av}$and T_f ,we determine Z_{av}.

Integration

$$-\int_{P_1}^{P_2} \frac{P}{Z_{av}}\, dP = \frac{8fMT_f P_{sc}^2 Q_{sc}^2}{R\pi^2 T_{sc}^2 g d^5} \int_0^L d\,L$$

After Integration

$$\frac{P_1^2 - P_2^2}{2} = \frac{8fMT_f Z_{av} L P_{sc}^2 Q_{sc}^2}{R\pi^2 T_{sc}^2 g d^5}$$

$$P_1^2 - P_2^2 = \left(\frac{16}{\pi^2}\right)\left(\frac{M}{R}\right)\left(\frac{P_{sc}}{T_{sc}}\right)^2 \frac{f\, T_f\, Z_a}{g\, d^5} Q_{sc}^2\, L$$

$$\Rightarrow Q_{sc}^2 = \left(\frac{T_{sc}}{P_{sc}}\right)^2 \left(\frac{P_1^2 - P_2^2}{f\, L\, M\, T_f\, Z_a}\right) d^5 \left(\frac{gR\pi^2}{16}\right)$$

Substituting gas properties

Taking, $M = 28.96\, G$; where, G= specific gravity (gas gravity)

$$g = 32.2\ ft/s^2$$

$$R = 1545\ \frac{ft \cdot lbf}{lb\text{–}mol \cdot {}^\circ R}$$

$$Q_{sc}^2 = \left(\frac{T_{sc}}{P_{sc}}\right)^2 \left(\frac{P_1^2 - P_2^2}{fLMT_fZ_a}\right) d^5 \left(\frac{gR\pi^2}{16}\right)$$

$M = 28.96\,G, g = 32.2, R = 1545$

$$Q_{sc}^2 = \left(\frac{T_{sc}}{P_{sc}}\right)^2 \left(\frac{P_1^2 - P_2^2}{fL(28.96G)T_fZ_a}\right) d^5 \left(\frac{gR\pi^2}{16}\right)$$

$$Q_{sc}^2 = \left(\frac{T_{sc}}{P_{sc}}\right)^2 \left\{\frac{(P_1^2 - P_2^2)d^5}{fLGT_fZ_a}\right\} \left(\frac{gR\pi^2}{16 \times 28.96}\right)$$

substitute $g$and R

$$Q_{sc}^2 = \left(\frac{T_{sc}}{P_{sc}}\right)^2 \left\{\frac{(P_1^2 - P_2^2)d^5}{fLGT_fZ_a}\right\} \left(\frac{32.2 \times 1545 \times \pi^2}{16 \times 28.96}\right)$$

$$Q_{sc}^2 = 1059.7 \left(\frac{T_{sc}}{P_{sc}}\right)^2 \left\{\frac{(P_1^2 - P_2^2)d^5}{fLGT_fZ_a}\right\}$$

take square root

$$Q_{sc} = 32.55 \left(\frac{T_{sc}}{P_{sc}}\right) \left\{\frac{(P_1^2 - P_2^2)d^5}{fLGT_fZ_a}\right\}^{1/2}$$

$$Q_{sc} = 32.55\left(\frac{T_{sc}}{P_{sc}}\right)\left\{\frac{\left(P_1^2 - P_2^2\right)d^5}{fLGT_fZ_a}\right\}^{1/2}$$

- Q_{sc} in ft^3/s
- T_{sc}, T_f in $°R$
- P_{sc}, P_1, P_2 in lbf/ft^2
- d in ft
- L in ft
- f dimensionless
- G dimensionless
- Z_a dimensionless
- $g = 32.2\ ft/s^2$
- $R = 1545\ \frac{ft \cdot lbf}{lb-mol \cdot °R}$

Gas Pipeline Flow Equation

$$Q_{sc} = 77.28\left(\frac{T_{sc}}{P_{sc}}\right)\left\{\frac{\left(P_1^2 - P_2^2\right)\, d^5}{f\ L\ G\ T_f\ Z_a}\right\}^{1/2} \text{----------}(iv)$$

Where,

- Q_{sc} =Gas flow rate in SCF/day at $T_{sc}$and P_{sc}
- T_{sc} =Base or standard temperature = 520°R
- P_{sc} =Base or standard pressure = 14.7 psia
- $P_1$and P_2 =Inlet and outlet pressures in psia
- d= Internal diameter of pipe in inches
- L= Length of pipeline in miles
- G= Gas gravity (air = 1)
- T_f =Gas flowing temperature or average line temperature (°R)
- f= Friction factor
- Z_{av} =Average gas compressibility factor

For Weymouth Equation we assume

$$f = \frac{0.032}{d^{1/3}}, d \ is \ in \ inches$$

Putting in equation $(\boldsymbol{iv})$ we get:

Weymouth Equation

$$\boldsymbol{Q_{sc} = 433.49 \left(\frac{T_{sc}}{P_{sc}}\right) \left[\frac{\left(P_1^2 - P_2^2\right)}{L\ G\ T_f\ Z_a}\right]^{1/2} d^{8/3}} \quad ---------- \ (\boldsymbol{v})$$

Validity

Size of pipe **within 12 inches** (gives very accurate result).

Question:

Natural gas flows through a horizontal pipeline under isothermal conditions. The following data are given:

- Inlet pressure, $P_1 = 900\ psia$
- Outlet pressure, $P_2 = 700\ psia$
- Pipe length, $L = 50\ miles$
- Internal diameter, $d = 10\ in$
- Base pressure, $P_{sc} = 14.7\ psia$
- Base temperature, $T_{sc} = 520°R$
- Flowing gas temperature, $T_f = 530°R$
- Gas gravity, $G = 0.65$
- Average compressibility factor, $Z_a = 0.90$

For the general gas flow equation, take friction factor, $f = 0.015$

Using the above data:

(a) Calculate the gas flow rate Q_{sc} in MMSCF/day using the general gas flow equation

(b) Calculate the gas flow rate Q_{sc} in MMSCF/day using the Weymouth equation

(c) Compare the two answers.

Solution:

General Equation (a)

$$P_1^2 - P_2^2 = 900^2 - 700^2$$
$$= 810000 - 490000$$
$$= 320000$$

$$Q_{sc} = 77.28\left(\frac{T_{sc}}{P_{sc}}\right)\left[\frac{(P_1^2 - P_2^2)d^5}{fLGT_fZ_a}\right]^{1/2}$$

$$\frac{T_{sc}}{P_{sc}} = \frac{520}{14.7} = 35.374$$

$$d^5 = 10^5 = 100000$$
$$fLGT_fZ_a = 0.015 \times 50 \times 0.65 \times 530 \times 0.90 = 232.5375$$
$$\frac{(P_1^2 - P_2^2)d^5}{fLGT_fZ_a} = \frac{320000 \times 100000}{232.5375}$$
$$\approx 1.3761 \times 10^8$$
$$(1.3761 \times 10^8)^{1/2} \approx 11730.8$$
$$Q_{sc} = 77.28 \times 35.374 \times 11730.8$$
$$Q_{sc} \approx 32065000\ SCF/day$$
$$\boxed{Q_{sc} \approx 3.21 \times 10^7\ SCF/day}$$
$$\boxed{Q_{sc} \approx 32.1\ MMSCFD}$$

Weymouth Equation (b)

$$Q_{sc} = 433.49\left(\frac{T_{sc}}{P_{sc}}\right)\left[\frac{(P_1^2 - P_2^2)}{LGT_f Z_a}\right]^{1/2} d^{8/3}$$

$$LGT_f Z_a = 50 \times 0.65 \times 530 \times 0.90$$

$$= 15502.5$$

$$\frac{(P_1^2 - P_2^2)}{LGT_f Z_a} = \frac{320000}{15502.5}$$

$$\approx 20.642$$

$$\sqrt{20.642} \approx 4.543$$

$$d^{8/3} = 10^{8/3}$$

$$= 10^{2.6667} \approx 464.16$$

$$Q_{sc} = 433.49 \times 35.374 \times 4.543 \times 464.16$$

$$Q_{sc} \approx 32390000\ SCF/day$$

$$\boxed{Q_{sc} \approx 3.24 \times 10^7\ SCF/day}$$

$$\boxed{Q_{sc} \approx 32.4\ MMSCFD}$$

(c) Comparison

From general equation:

$$Q_{sc} \approx 32.1\ MMSCFD$$

From Weymouth equation:

$$Q_{sc} \approx 32.4\ MMSCFD$$

Difference

$$\text{Difference} = 32.4 - 32.1 = 0.3\ MMSCFD$$

The two values are very close.

California Natrual Gas Association (CNGA) method

This equation is used for calculating the compressibility factor (Z) when the gas gravity, temperature, and pressure are known.

$$Z = \frac{1}{\left[1 + \left(\frac{P_{avg}\ 344,400\ (10)^{1.785G}}{T_f^{3.825}}\right)\right]}$$

Where:

P_{avg} =Average pressure, psig

T_f =Average gas temperature, °R

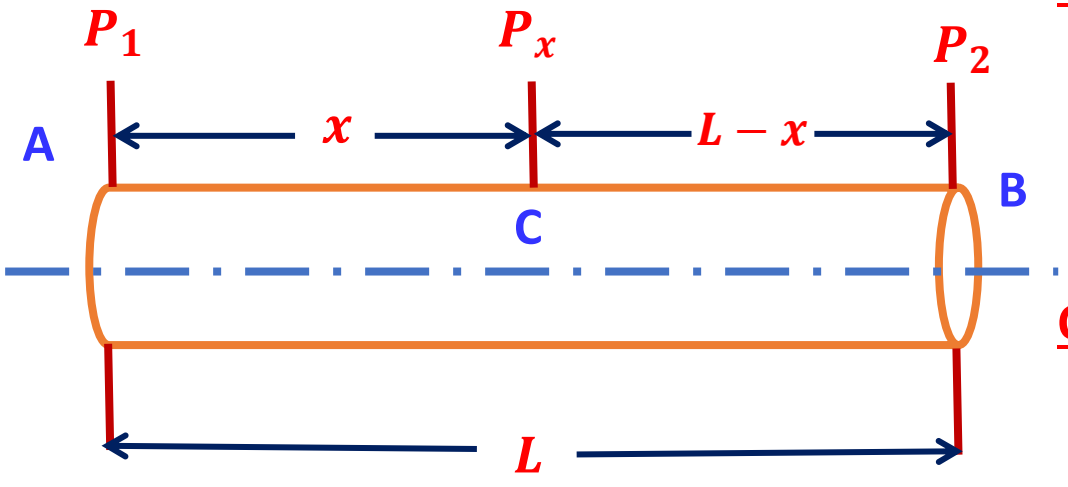
G= Gas gravity (air = 1)

This formula for compressibility factor is valid when the average gas pressure P_{avg} is greater than 100 psig.

For pressure less than 100 psig, the compressibility factor is approximately:

$$Z \approx 1.0$$

Average Pressure in Gas pipeline



A to C

$$Q = 77.28\left(\frac{T_s}{P_s}\right)\left(\frac{P_1^2 - P_x^2}{f x G T_f Z_a}\right)^{1/2} d^{5/2} \text{ ---------- } (i)$$

C to B

$$Q = 77.28\left(\frac{T_s}{P_s}\right)\left(\frac{P_x^2 - P_2^2}{f(\mathrm{L} - x) G T_f Z_a}\right)^{1/2} d^{5/2} \text{ ---- } (ii)$$

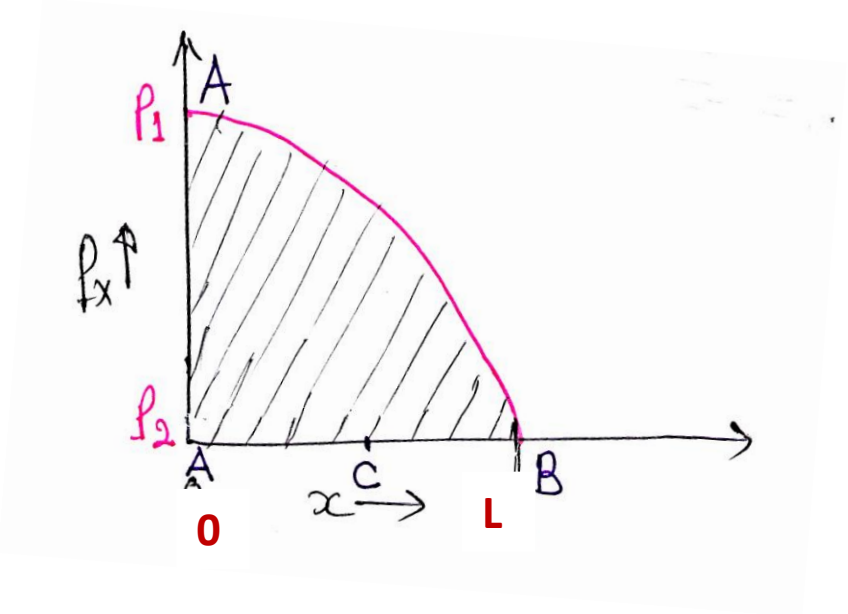
$$P_{avg} = P_m = \frac{2}{3}\left[\frac{P_1^3 - P_2^3}{P_1^2 - P_2^2}\right]$$

$$\frac{P_1^2 - P_x^2}{x} = \frac{P_x^2 - P_2^2}{(\mathrm{L} - x)}$$

$$P_x{}^2 = P_1{}^2 - \frac{x(P_1{}^2 - P_2{}^2)}{L}$$

$$P_x = \left[P_1^2 - \frac{x(P_1^2 - P_2^2)}{L}\right]^{1/2} \text{ ---------- } (iii)$$

Average Pressure in Gas pipeline



$$\int (ax+b)^n dx$$

$$= \frac{1}{a}\frac{(ax+b)^{n+1}}{n+1}$$

$$P_{avg} = \frac{1}{L}\int_0^L P_x \, dx$$

$$= \frac{1}{L}\int_0^L \left[P_1^2 - \frac{x(P_1^2 - P_2^2)}{L}\right]^{1/2} dx$$

Let

$$A = P_1^2, B = \frac{P_1^2 - P_2^2}{L}$$

Then the integral becomes

$$\frac{1}{L}\int_0^L \sqrt{A - Bx} \, dx$$

Now use the standard formula

$$\int \sqrt{a - bx} \, dx = -\frac{2}{3b}(a - bx)^{3/2}$$

So,

$$\frac{1}{L}\int_0^L \sqrt{A - Bx} \, dx = \frac{1}{L}\left[-\frac{2}{3B}(A - Bx)^{3/2}\right]_0^L$$

Average Pressure in Gas pipeline

Now substitute limits:

At $x = L$,

$$A - BL = P_1^2 - \left(\frac{P_1^2 - P_2^2}{L}\right) L = P_2^2$$

$$\left(A - BL\right)^{3/2} = \left(P_2^2\right)^{3/2} = P_2^3$$

At $x = 0$,

$$A^{3/2} = \left(P_1^2\right)^{3/2} = P_1^3$$

Thus,

$$\frac{1}{L}\int_0^L \sqrt{A - Bx}\ dx = \frac{1}{L}\left[-\frac{2}{3B}(P_2^3 - P_1^3)\right]$$

$$= \frac{2}{3BL}(P_1^3 - P_2^3)$$

Since

$$B = \frac{P_1^2 - P_2^2}{L} \Rightarrow BL = P_1^2 - P_2^2$$

therefore,

$$\boxed{\frac{1}{L}\int_0^L \left[P_1^2 - \frac{x(P_1^2 - P_2^2)}{L}\right]^{1/2} dx = \frac{2}{3}\ \frac{P_1^3 - P_2^3}{P_1^2 - P_2^2}}$$

Storage Capacity of Gas pipeline

Storage capacity of a pipeline is defined as the difference between the gas content under packed (off-peak period) and unpacked (peak period) condition.

A pipeline is said to be packed when the withdrawal from the pipeline is minimum, and therefore for a constant supply of gas, the discharge pressure is maximum.

ΔP is minimum

Similarly, a pipeline is said to be unpacked when withdrawal is maximum, and for a constant supply of gas the discharge pressure is minimum.

ΔP is maximum

Gas content

$$V = 1.955\, d^2\, L\, P_m\, K$$

Where:

V= Gas content (ft^3)

$d = \mathrm{I.D(inch)}$

P_m = Mean pressure (psi)

$$\boldsymbol{P_m = \frac{2}{3}\left(\frac{P_1^3 - P_2^3}{P_1^2 - P_2^2}\right)}$$

L =Length of line (miles)

K= Super compressibility factor

$$K = \left(\frac{1}{Z}\right)^{1/2}$$

Where, Z= Compressibility factor.

$$\textbf{Storage } \boldsymbol{Capacity} = \boldsymbol{V_{packed} - V_{unpacked}}$$

How to calculate Za by Standing and Kats Compressibility Factor Chart

$$P_{avg} = \frac{2}{3}\left[\frac{P_1^{\,3} - P_2^{\,3}}{P_1^{\,2} - P_2^{\,2}}\right]$$

For given gas gravity (γ_g), find pseudocritical pressure (P_c) and pseudocritical temperature (T_c)

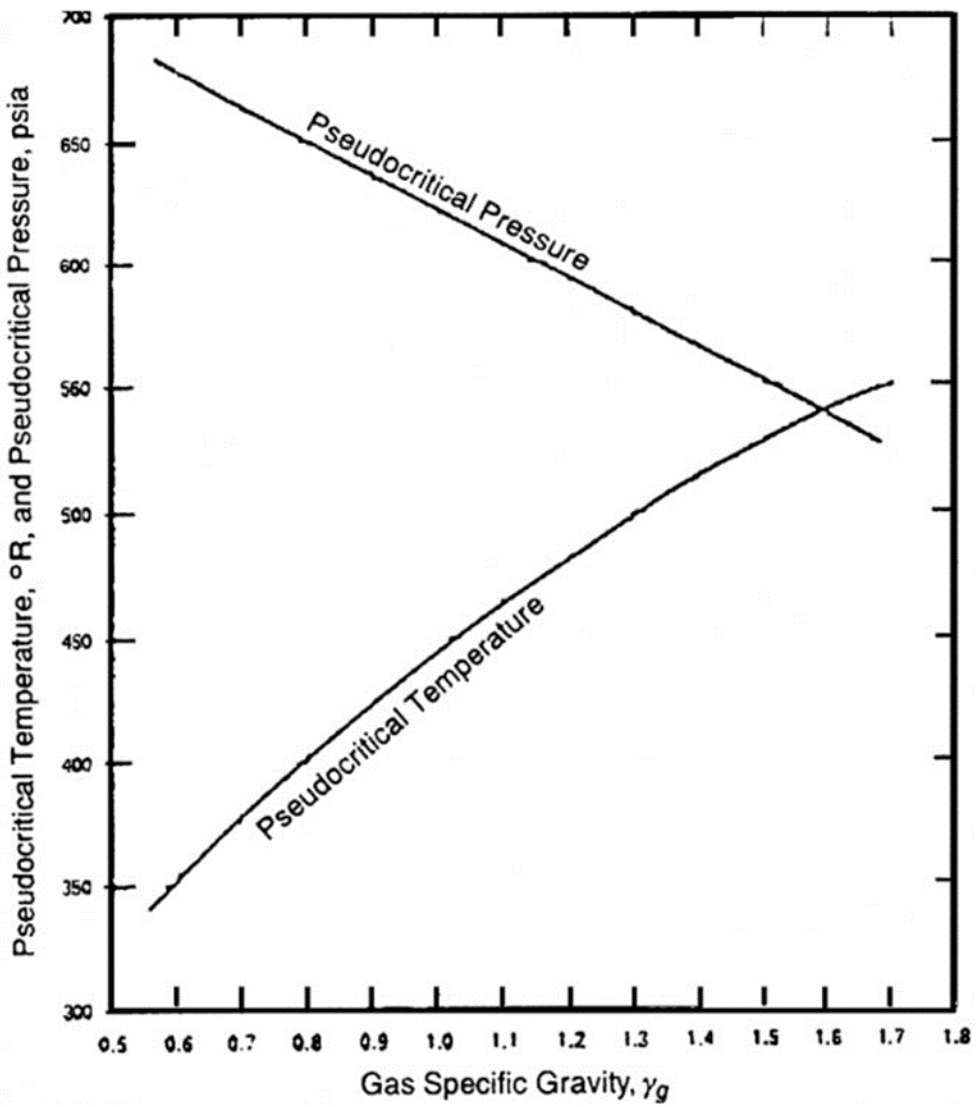


Fig.: Pseudo critical properties chart (Brown et al 1948)

Standing-Katz chart or gas Z-factor correlation

Step 1: Find average pressure P_{av}

$$P_{av} = \frac{2}{3}\left(\frac{P_2^3 - P_1^3}{P_2^2 - P_1^2}\right)$$

Step 2: Take flowing temperature

If flow is isothermal, then

$$T_f = constant$$

If temperature is not constant, use an average temperature

$$T_{av} = \frac{T_1 - T_2}{\ln(T_1/T_2)}$$

Step 3: Find pseudo-reduced properties

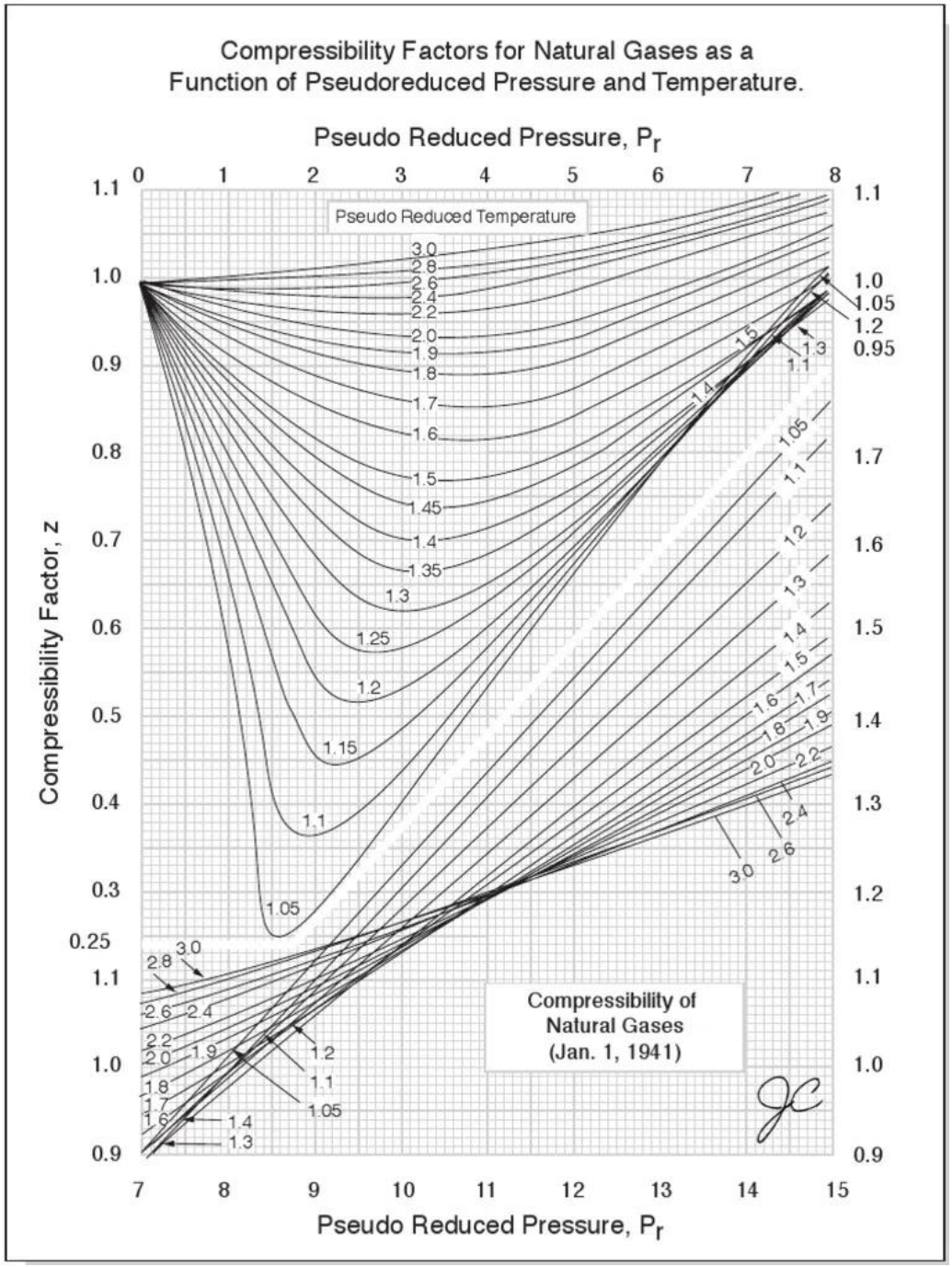
To calculate Z, first find:

$$P_{pr} = \frac{P_{av}}{P_{pc}}, \qquad T_{pr} = \frac{T_f}{T_{pc}}$$

where

- ❖ P_{pc} =pseudo-critical pressure
- ❖ T_{pc} =pseudo-critical temperature

Step 4: Use P_{pr} and T_{pr} to get Z_{av}



Ques: A pipeline 200 miles long 19.250" I.D. is operated under conditions given below:-

Gas Temperature 60°F, S.G. = 0.65

At period of low flow:

$P_1 = 800$ psia, $P_2 = 600$ psia, $k = 1.087$

At period of high flow:

$P_1 = 800$ psia, $P_2 = 400$ psia, $k = 1.075$

Determine the storage capacity of line.

Solution

In Low flow:

$$P_m = \frac{2}{3}\left(\frac{P_1^3 - P_2^3}{P_1^2 - P_2^2}\right)$$
$$= \frac{2}{3}\left(\frac{800^3 - 600^3}{800^2 - 600^2}\right)$$
$$P_m = 704.76\ psi$$

In high flow:

$$P_m = \frac{2}{3}\left(\frac{800^3 - 400^3}{800^2 - 400^2}\right)$$
$$P_m = 622.22\ psi$$

$$V_{packed} = 1.955 \times (19.250)^2 \times 200 \times 704.76 \times 1.08$$
$$V_{packed} = 1{,}109{,}964.31\ ft^3$$

$$V_{unpacked} = 1.955 \times (19.250)^2 \times 200 \times 622.22 \times 1.075$$
$$= 9{,}691{,}4923\ ft^3$$

$$\text{Storage } Capacity = V_{packed} - V_{unpacked}$$
$$= 110996431 - 96914923$$
$$\text{Storage } Capacity = 14{,}081{,}508\ ft^3 \approx 14.08\ MMcft$$